



K-Feldspar Megacryst Fabrics and Crystal Size Distributions as Constraints on the Synmagmatic Emplacement of the Abeokuta Porphyritic Granite, Southwestern Nigeria

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Abstract: The Abeokuta porphyritic granite of southwestern Nigeria contains abundant K-feldspar megacrysts whose morphology and spatial organization preserve information on crystallization and magma emplacement. In this study, the development of the K-feldspar megacrystic fabric and its relationship to magmatic evolution and emplacement were evaluated by integrating field-based morphometric measurements, preferred crystal orientation, petrographic observations, crystal size distribution (CSD) analysis, and regional geological constraints. Approximately 1,500 K-feldspar megacrysts were measured on exposed horizontal outcrop surfaces for length, width, and long-axis orientation, with partially exposed and truncated crystals excluded. Corrected morphometric analysis yielded a two-dimensional shape anisotropy ratio of 1.575 and a preferred long-axis orientation of approximately 054°, defining a NE–SW-trending magmatic fabric. CSD analysis was performed on five representative K-feldspar populations using CSDCorrections v1.4.0, with 42–62 crystals analyzed per sample. Regression slopes ranged from -0.414 to -0.811 mm^{-1} , with intercepts ranging from -3.01 to -4.84 mm^{-4} ; the pooled population yielded a slope of -0.587 mm^{-1} and an intercept of -4.11 mm^{-4} . Variations in slope and departures from linearity indicate that the K-feldspar populations were not produced under a single, invariant nucleation-and-growth regime and are instead consistent with temporally evolving crystallization and post-nucleation textural modification, including crystal growth and coarsening. The preferred orientation of the K-feldspar megacrysts is interpreted predominantly as a synmagmatic fabric generated during magma emplacement and flow, although subordinate modification during subsequent solid-state deformation cannot be excluded. Regional geochemical studies provide a broader Pan-African petrogenetic framework but cannot be regarded as direct geochemical evidence for the Abeokuta granite. Accordingly, crust-mantle interaction, magma recharge or mixing, assimilation, and hybrid magma sources remain plausible but untested petrogenetic hypotheses. These results demonstrate that integrated quantitative analysis of K-feldspar megacryst fabrics and CSDs can constrain the crystallization history and emplacement dynamics of porphyritic granitic magmas within Pan-African crust.

Keywords: Pan-African orogeny; Strain analysis; Porphyritic granite; Megacryst; Magma evolution; Abeokuta granite; Crystal size distribution

1 Introduction

Granitoid plutons preserve important records of magma generation, crystallization, transport, and emplacement within the continental crust. Their mineral textures and internal fabrics can provide information about the prevailing physical conditions during crystallization and the processes that operated as magma evolved from a mobile melt to a crystalline rock [1]. Among these textural features, large feldspar crystals are particularly useful because their size, morphology, and spatial organization can preserve information about crystal growth and magma dynamics. Therefore, quantitative characterization of feldspar populations provides an effective approach for investigating the relationship between crystallization and emplacement in porphyritic granitoids [2]. The development of large K-feldspar megacrysts in granitic rocks may involve prolonged crystal growth, changes in nucleation and growth rates, crystal-melt interaction, and late-stage textural coarsening. Consequently, megacryst size alone does not

necessarily represent a single stage of magmatic evolution. Crystal size distribution (CSD) analysis provides a quantitative framework for evaluating such crystal populations by relating crystal abundance to crystal size and thereby assessing variations in nucleation, growth, and textural modification. Deviations from simple CSD relationships may arise from changing crystallization conditions, crystal accumulation or removal, mixing of crystal populations, or textural coarsening. Consequently, CSD analysis is most informative when integrated with crystal morphology and petrographic observations [3, 4]. Crystal orientation provides a complementary record of magma dynamics. Crystals suspended in a mobile magma may rotate and become preferentially organized during flow, producing a magmatic fabric that can be preserved after solidification. In granitoids containing abundant feldspar megacrysts, preferred crystal orientation can therefore provide information about magma movement and emplacement. However, crystal alignment may develop before complete crystallization and should consequently be distinguished from the fabrics produced by post-solidification deformation. The combined analysis of crystal morphology, preferred orientation, and CSD offers a means of examining both the spatial organization and crystallization history of a crystal population [2, 5].

The Nigerian Basement Complex forms part of the Pan-African mobile belt of West Africa and contains a diverse assemblage of high-grade metamorphic rocks, migmatites, metasedimentary and metavolcanic rocks, and granitoid intrusions. Southwestern Nigeria contains extensive Pan-African granitoids emplaced within a structurally complex basement terrain. These granitoids provide important records of Neoproterozoic crustal evolution, deformation, and magmatism. The Abeokuta area of southwestern Nigeria forms part of this basement province and contains several granitoid lithologies, including biotite granite and porphyritic granite, together with migmatite-gneiss and granite-gneiss [6]. The Abeokuta porphyritic granite is particularly distinctive because of its abundant K-feldspar megacrysts. The crystals occur as prominent euhedral to subhedral grains within a medium- to coarse-grained quartz-feldspar-biotite matrix and locally display perthitic intergrowths and zoning. Their conspicuous size and abundance make the granite an appropriate natural laboratory for investigating the relationship between feldspar crystallization, crystal organization, and emplacement [7]. The broader structural framework of southwestern Nigeria is characterized by several dominant orientations, including NE-SW, NNE-SSW, N-S, and E-W trends. Such structural anisotropies may influence magma ascent and emplacement by providing preferential pathways through the basement. Within the Abeokuta area, regional lineaments include a significant NE-SW component. The relationship between these regional structural trends and the orientation of K-feldspar megacrysts provides an opportunity to examine whether the regional structural architecture is reflected in the internal magmatic fabric of the porphyritic granite [8, 9].

Despite the importance of K-feldspar megacrysts to the textural character of the Abeokuta porphyritic granite, quantitative constraints on their morphology, preferred orientation, and CSD remain limited. In particular, the relationship between K-feldspar crystal shape, directional organization, and crystal population evolution has not been adequately integrated into a single emplacement model. Therefore, a quantitative approach is required to determine whether the megacrysts form a randomly distributed population or possess a systematic fabric, and to establish whether their size distribution is compatible with a simple crystallization history or changing crystal-growth conditions. Therefore, the principal research problem addressed in this study is the lack of an integrated quantitative assessment of the K-feldspar population of the Abeokuta porphyritic granite. Specifically, it is necessary to establish the degree of crystal shape anisotropy, determine whether the megacrysts possess a preferred orientation, characterize their CSD, and evaluate the significance of these characteristics for the crystallization and emplacement history of the granite. This study addresses these questions through field-based measurements of K-feldspar megacryst dimensions and long-axis orientations, quantitative morphometric analysis, petrographic characterization, and CSD modeling. Approximately 1,500 K-feldspar megacrysts were measured on exposed outcrop surfaces, while representative K-feldspar populations were analyzed using CSDCorrections v1.4.0 to obtain stereologically corrected CSDs. Regional lineament analysis provides an independent structural framework within which the crystal fabric can be evaluated. Therefore, the aim of this study is to constrain the crystallization and synmagmatic emplacement history of the Abeokuta porphyritic granite through quantitative analysis of K-feldspar megacryst morphology, preferred orientation, and CSD.

The specific objectives are to characterize the morphology and two-dimensional shape anisotropy of K-feldspar megacrysts from field-based length and width measurements; determine their preferred long-axis orientation and resulting crystal fabric; quantify CSDs of representative K-feldspar populations using stereological correction and CSD modeling; assess how these distributions vary with changing crystallization and textural conditions; integrate morphology, preferred orientation, CSD, and regional structural data to constrain granite emplacement; and situate the findings within the broader Pan-African tectonomagmatic framework of southwestern Nigeria. The central hypothesis tested is that the K-feldspar megacryst population preserves complementary records of magmatic crystal organization and crystallization history, such that its preferred orientation and CSD characteristics can be used together to constrain the emplacement and textural evolution of the Abeokuta porphyritic granite [2, 10].

The principal contribution of this study is the integration of K-feldspar morphometry, preferred crystal orientation,

and CSD analysis within a single framework for interpreting Pan-African porphyritic granite. This approach provides a quantitative basis for distinguishing crystal population characteristics associated with magmatic organization and evolving crystallization conditions from those that would require independent evidence of post-solidification deformation or specific magma-source processes.

2 Geological Setting

2.1 Regional Geological Framework and Local Geology of the Abeokuta Area

The study area is located in the Abeokuta region of southwestern Nigeria, within the southwestern sector of the Nigerian Basement Complex. The Nigerian Basement Complex is part of the Pan-African mobile belt of West Africa and preserves a complex history of Precambrian crustal deformation, metamorphism, migmatization, and granitoid magmatism [11]. The Pan-African tectonothermal episode produced widespread granitoid emplacement and structural reworking across the region, resulting in a heterogeneous basement assemblage comprising gneisses, migmatites, granitoids, and metasedimentary–metavolcanic units. The Abeokuta area forms part of this Pan-African granitoid province and contains several granitoid bodies emplaced within the older basement assemblage. The regional geological framework is characterized by structural heterogeneity, with granitoid bodies occurring in association with metamorphic basement rocks and regional structural fabrics [12, 13]. Therefore, this area provides an appropriate setting for investigating the relationship between granitoid crystallization, internal crystal fabric, and emplacement within a structurally complex basement [14] (Figure 1).

The local geology comprises high-grade basement lithologies and Pan-African granitoids. The principal lithological units recognized in the study area include migmatite-gneiss, granite-gneiss, biotite granite, porphyritic granite, and subordinate fine-grained granite. Migmatite-gneiss and granite-gneiss are important components of the basement framework, whereas granitoid bodies occur as intrusive units within this crystalline terrain (Figure 1). The porphyritic granite is distinguished by its conspicuous K-feldspar megacrysts and occurs as isolated hills, ridges, and low-lying outcrops. The porphyritic granite is medium- to coarse-grained and contains large K-feldspar crystals within a quartz–feldspar–biotite matrix. The megacrysts are commonly euhedral to subhedral and sufficiently well exposed at several localities to permit direct measurement of their dimensions and long-axis orientations [7]. The geological relationships among the principal lithological units are illustrated in Figure 2. The detailed geological map identifies the distribution of porphyritic granite relative to the surrounding basement lithologies and provides a spatial framework for sampling and structural observations.

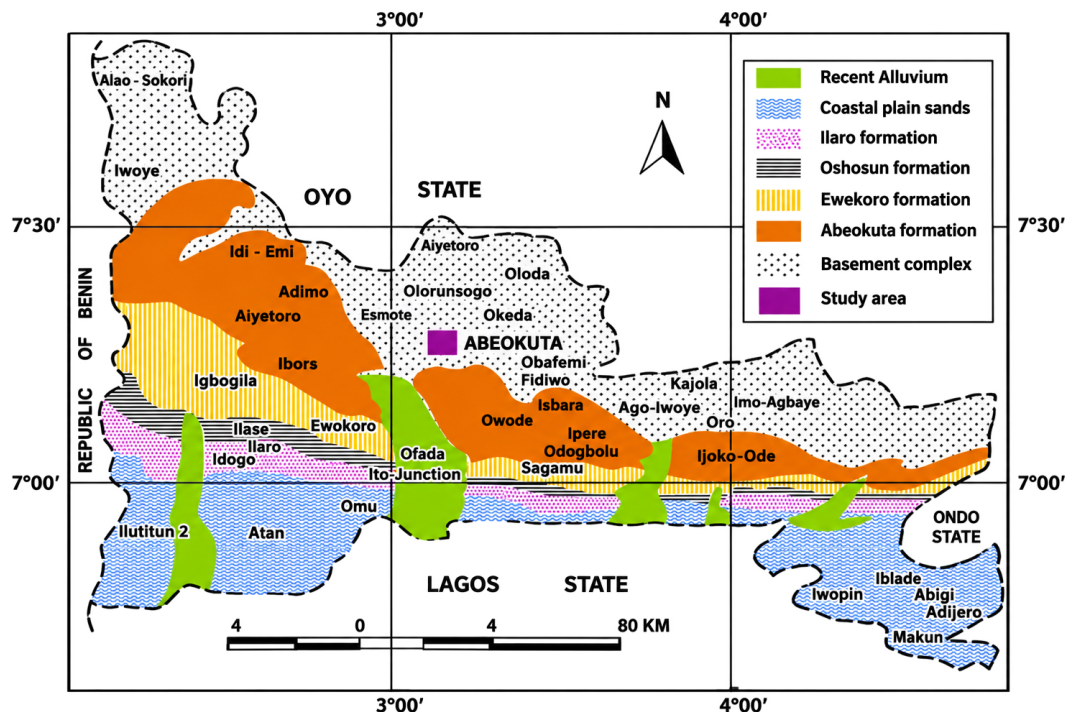


Figure 1. Regional geologic map of the study area

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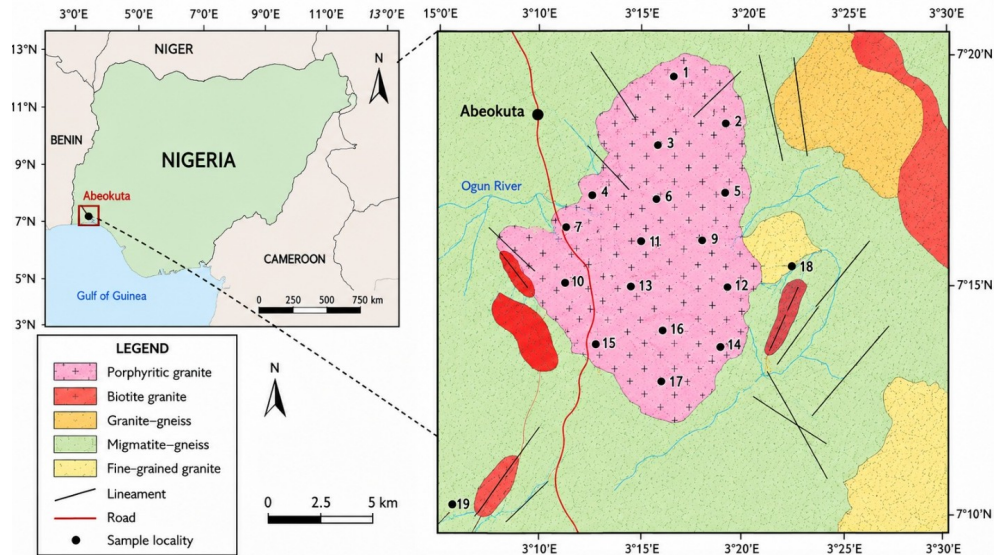


Figure 2. Geological setting of the Abeokuta study area showing the distribution of the porphyritic granite and associated basement lithologies, major structural features, and sampling localities

2.2 Structural Framework

The southwestern Nigerian Basement Complex is characterized by prominent structural fabrics with several recurring orientations. Within the Abeokuta region, the regional lineaments include the N–S, E–W, and NE–SW trends [8]. Remote-sensing analysis of the Abeokuta SE Sheet 260 identified a substantial population of lineaments, with the overall mean orientation trending approximately NE–SW. The regional structural framework provides an important context for granitoid emplacement because fractures, shear zones, and other crustal discontinuities can influence magma migration and emplacement [16]. In the Abeokuta area, the distribution of lineaments is spatially heterogeneous, with greater lineament concentrations in the basement complex terrain than in the adjoining sedimentary domains. The detailed study area contains a smaller number of mapped lineaments, concentrated principally around the flanks of the Ogun River. In this research, the structural framework is considered as a regional geological constraint. The Results section quantitatively evaluates the relationship between regional lineament orientations and measured K-feldspar fabric orientations to assess their potential structural correspondence and implications for magma emplacement.

2.3 Pan-African Magmatic Context

The granitoid rocks of southwestern Nigeria are part of the broader Pan-African magmatic assemblage of the West African Mobile Belt. The Pan-African event involved widespread deformation, metamorphism, and granitoid emplacement, producing a heterogeneous association of syn- and post-tectonic granitoids together with reworked older basement rocks [17, 18]. The Abeokuta porphyritic granite is considered within the regional Pan-African framework. Its occurrence within a structurally complex basement, together with the presence of a measurable K-feldspar fabric, makes it particularly suitable for examining the relationship between crystal growth and magma emplacement in the basement. The precise temporal relationship between granite emplacement and individual deformation phases cannot be established solely from the field fabric investigated in this study. Consequently, the regional Pan-African age framework was used to establish the tectonomagmatic setting, while the relative timing of crystal organization and solidification was assessed using petrographic, morphometric, and CSD data [1].

2.4 Regional Lineament Architecture

The regional lineament distribution (Figure 3) provides an additional structural perspective of the Abeokuta area. The analysis of the Abeokuta SE Sheet 260 identified 1,541 lineaments, with only nine falling within the detailed study area. The lineament density increased toward the northern and northeastern portions of the sheet, particularly within the basement-complex terrain. The frequency distribution is dominated by the N–S and E–W orientations, whereas the mean orientation of the regional lineament population trends is approximately NE–SW. This regional NE–SW component is significant in the present study because the K-feldspar megacrysts also exhibit a preferred NE–SW orientation. The geometric relationship between the two datasets was investigated without assuming that the crystal orientation represented a particular principal stress direction.

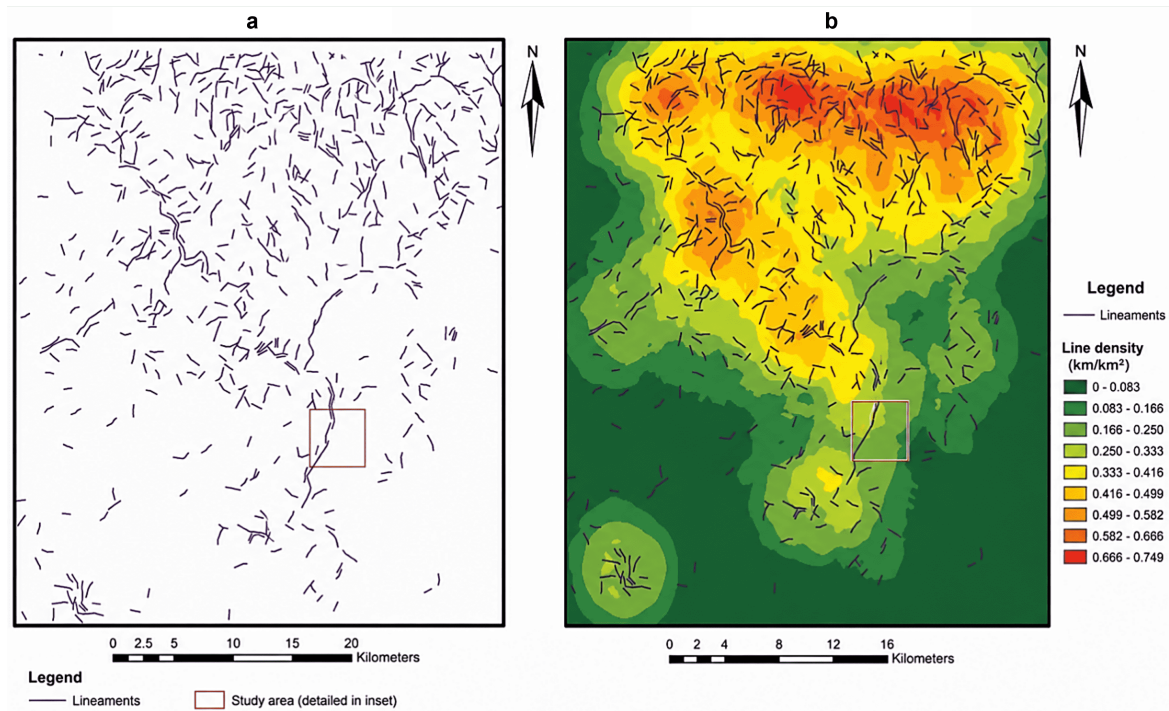


Figure 3. Regional lineament analysis of the Abeokuta area: (a) extracted lineament map; (b) lineament density distribution (km/km²)

3 Materials and Methods

3.1 Study Design and Analytical Workflow

This study employed an integrated approach combining field observations, petrographic characterization, morphometric analysis, CSD analysis, and remote sensing to characterize the K-feldspar population of the Abeokuta porphyritic granite. The workflow comprised the following steps: (i) field reconnaissance and geological observation; (ii) systematic measurement of K-feldspar megacryst dimensions and orientations; (iii) petrographic characterization of the granite; (iv) quantitative analysis of crystal morphology and shape anisotropy; (v) CSD analysis and stereological conversion of two-dimensional measurements to three-dimensional crystal populations; and (vi) regional lineament analysis. The analytical framework was designed to distinguish directly measured crystal characteristics from subsequent geological interpretation. Therefore, field measurements were treated as two-dimensional observations of crystals exposed on outcrop surfaces, whereas the CSD analysis incorporated stereological correction to estimate three-dimensional CSDs. This approach allows the morphology, orientation, and crystal size characteristics of the K-feldspar population to be evaluated independently before their integration with the regional structural framework.

3.2 Field Investigation and Sampling Strategy

Fieldwork was conducted at 19 localities within the Abeokuta porphyritic granite. Exposures were accessible, fresh granite surfaces where K-feldspar megacrysts could be distinguished from the groundmass. The population was examined directly on outcrop surfaces rather than from thin sections or polished slabs. Measurements were taken on horizontal surfaces, providing a consistent reference plane for orientation. Partially exposed, broken, truncated, weathered, or obscured crystals were excluded because reliable dimensions and measurable axes could not be established. Approximately 1,500 K-feldspar megacrysts were measured across 19 localities. Because smaller crystals were harder to distinguish from the groundmass, measurements focused on crystals with clear boundaries.

Measurements were made on exposed horizontal outcrop surfaces using a ruler and compass. For each K-feldspar megacryst, long-axis length (L), short-axis width (W), and long-axis azimuth were recorded. The long axis was the longest measurable dimension, and width was measured perpendicular to it across the widest part. Azimuth was measured relative to geographic north, providing a reference frame for comparing orientations among localities. Field photographs documented the exposed crystal populations and spatial organization (Figure 4 and Figure 5) and confirmed the measured populations, but were not the primary source of dimensional measurements. Thus, measurements represent two-dimensional intersections of three-dimensional crystals, and length and width were not interpreted as true three-dimensional dimensions.



Figure 4. Representative field photograph of the Abeokuta porphyritic granite showing conspicuous K-feldspar megacrysts within the medium- to coarse-grained matrix: (a) the intergrowth of K-feldspar megacrysts; (b) the megacrysts aligned in a preferred orientation

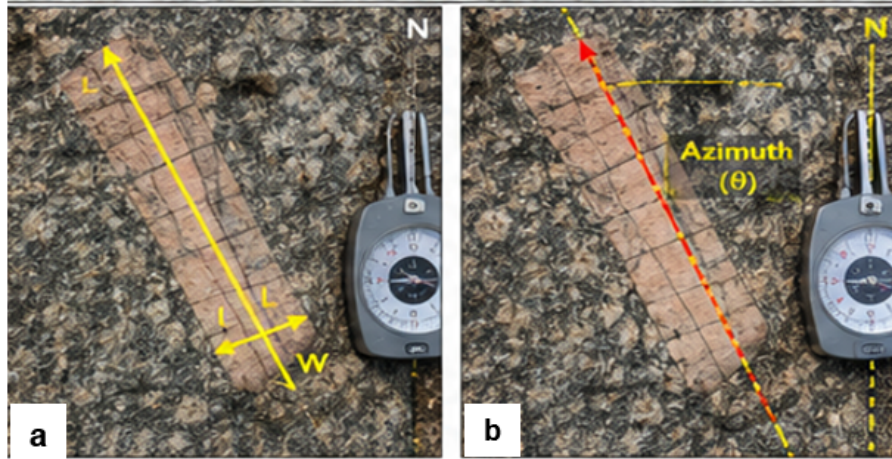


Figure 5. Field measurement procedure for K-feldspar megacrysts: (a) measurement of long-axis length (L) and short-axis width (W); (b) measurement of long-axis azimuth relative to the geographic north on the horizontal measurement surface

3.3 Data Screening and Quality Control

The morphometric dataset was screened prior to statistical analysis. The final dataset was constrained such that $L \geq W$. Partially exposed and truncated crystals were excluded because incomplete crystal boundaries can systematically underestimate the crystal dimensions and introduce bias into both the aspect ratio and CSD calculations. The resulting dataset was subsequently used to calculate the crystal dimensions, aspect ratios, and orientation distributions.

3.4 K-Feldspar Aspect-Ratio Analysis

The two-dimensional aspect ratio of each measurable K-feldspar crystal was calculated as:

$$R_f = L/W \quad (1)$$

where, R_f is the aspect ratio, L represents the long-axis length, and W represents the short-axis width.

An R_f value of 1 represents an equant two-dimensional crystal, whereas increasing values indicate greater crystal elongation. The arithmetic mean aspect ratio was calculated from the corrected individual crystal ratios.

$$R_f = \frac{\sum_{i=1}^n R_{f,i}}{n} \quad (2)$$

where, $R_{f,i}$ represents the final axial ratio of the i -th K-feldspar megacryst (L/W); n represents the total number of K-feldspar megacrysts included in the calculation; and i represents the index identifying each individual K-feldspar megacryst, ranging from 1 to n .

The distribution of the aspect ratios was evaluated using frequency plots and histograms. The length–width relationships were also examined to characterize the morphology of the K-feldspar population. Because the measurements were made on exposed two-dimensional surfaces and the original three-dimensional crystal shapes were not independently known, the resulting R_f values were used as morphometric descriptors rather than as direct measurements of finite three-dimensional tectonic strain.

3.5 Shape-Anisotropy and K-Feldspar Orientation

The corrected shape anisotropy parameter for the K-feldspar population was calculated independently from the morphometric dataset. Shape anisotropy quantified directional anisotropy within measured crystals and was considered independently of the arithmetic mean (L/W) ratio, as the two parameters describe distinct characteristics and are not interchangeable [19]. Azimuths of K-feldspar long axes were grouped into directional classes and plotted as rose diagrams to determine preferred orientation. The orientation distribution, analyzed from measured long-axis azimuths, identified a dominant orientation with the greatest concentration of measurements.

3.6 Crystal Size Distribution Analysis

CSD analysis characterized size-frequency relationships of the K-feldspar population and variations in crystal-population characteristics. Measurements of crystal intersections were used. Because measurements were two-dimensional, stereological correction was required before interpreting distributions as three-dimensional crystal populations. Five K-feldspar populations were selected. Measurable crystals ranged from 42 to 62 per population, with a minimum of 42 crystals. Thus, the analysis represents five K-feldspar populations rather than the approximately 1,500-crystal morphometric dataset. Two-dimensional crystal measurements were converted to three-dimensional CSDs using CSDCorrections v1.4.0. The software applies stereological corrections for relationships between two-dimensional intersections and three-dimensional populations, incorporating intersection probability and sectioning geometry. Measured crystal dimensions, fabric, shape, and stereological parameters were incorporated. Crystal population density was expressed as:

$$n(L) = \frac{dN(L)}{dL} \quad (3)$$

where, $N(L)$ is the cumulative number of crystals per unit volume with size $\leq L$, and $n(L)$ is the population density within a size interval.

The CSD relationships were evaluated on $\ln[n(L)]$ versus size L plots. Linear portions were fitted by regression to determine slope and intercept, reported in mm^{-1} and mm^{-4} . CSD profiles were examined for departures from linear behavior, including curvature, slope changes, and concave-downward or hump-shaped portions. Accordingly, CSD geometry was treated as an indicator of crystal population behavior rather than a diagnostic of a magmatic process.

3.7 Remote-Sensing and Lineament Analysis

Lineaments were identified from linear or curvilinear tonal, textural, and topographic features interpreted as geological discontinuities. Extracted features were digitized, and coordinates, lengths, and azimuths were recorded. Structural analysis of remote sensing data for Abeokuta SE Sheet 260 characterized lineament architecture and compared orientations with K-feldspar fabric. A total of 1,541 lineaments were identified, nine within the study area. The dataset was analyzed using orientation-frequency plots, rose diagrams, and lineament density to assess structural concentration and evaluate relationships between regional structures and the K-feldspar fabric. Subsequently, density and orientation-frequency products characterized the structural pattern. Principal parameters were L , W , L/W , R_s , azimuth, CSD slope, and CSD intercept [20, 21].

4 Results

4.1 Field and Petrographic Characteristics

The Abeokuta porphyritic granite occurs as isolated ridges and low-lying exposures within the Precambrian Basement Complex. The rock is medium- to coarse-grained and is characterized by conspicuous K-feldspar

megacrysts enclosed in a granular quartz–feldspar–biotite matrix. The megacrysts are predominantly euhedral to subhedral and locally display perthitic intergrowths and compositional zoning. The groundmass consisted mainly of quartz, feldspar, and biotite, with subordinate plagioclase and accessory zircon, apatite, and opaque minerals. Quantitative measurements were obtained from the exposed granitoid surfaces at 19 localities. K-feldspar was identified based on its characteristic optical properties and textural relationships with quartz, plagioclase, and biotite (Figure 6).

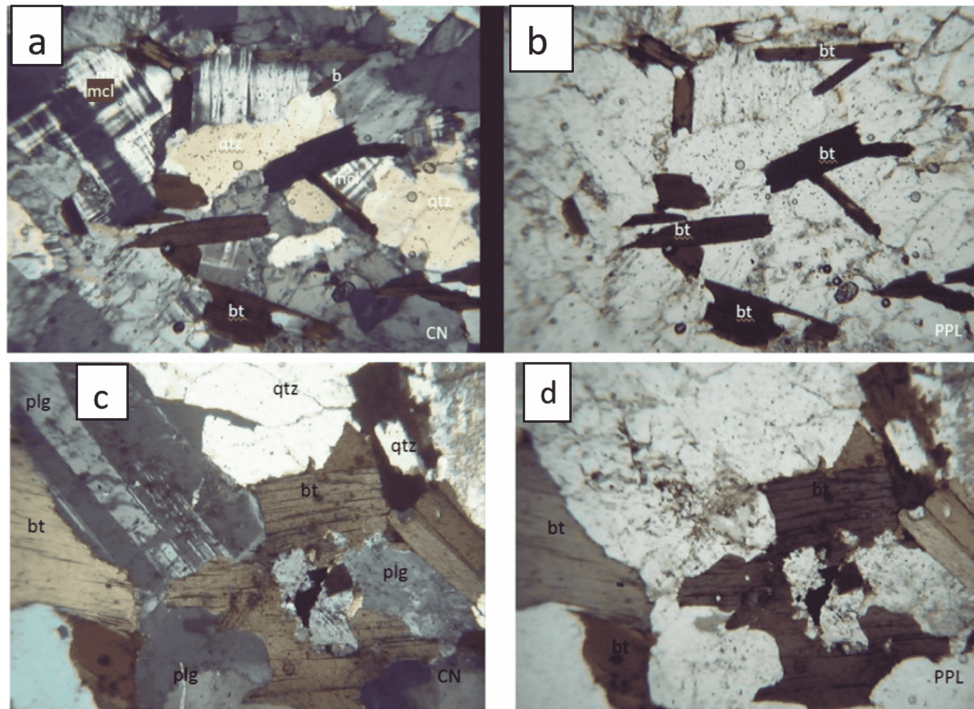


Figure 6. Representative photomicrographs of the Abeokuta porphyritic granite showing a K-feldspar megacryst and its textural relationships with quartz, plagioclase, and biotite: (a, c) crossed-polarized light; (b, d) plane-polarized light

4.2 K-Feldspar Megacryst Size and Morphology

Approximately 1,500 K-feldspar megacrysts were measured for their long-axis length (L), short-axis width (W), and long-axis orientation. The measurements represent two-dimensional intersections of three-dimensional crystals exposed on outcrop surfaces. The K-feldspar population displayed a broad range of crystal dimensions. The measured length and width distributions ranged from 2.5 to 6.7 cm and 1.2 to 3.5 cm, respectively. The corresponding mean dimensions were 4.56 cm in length and 2.34 cm in width, respectively. The resulting mean length-to-width ratio was $L/W = 1.95$ (Figure 7).

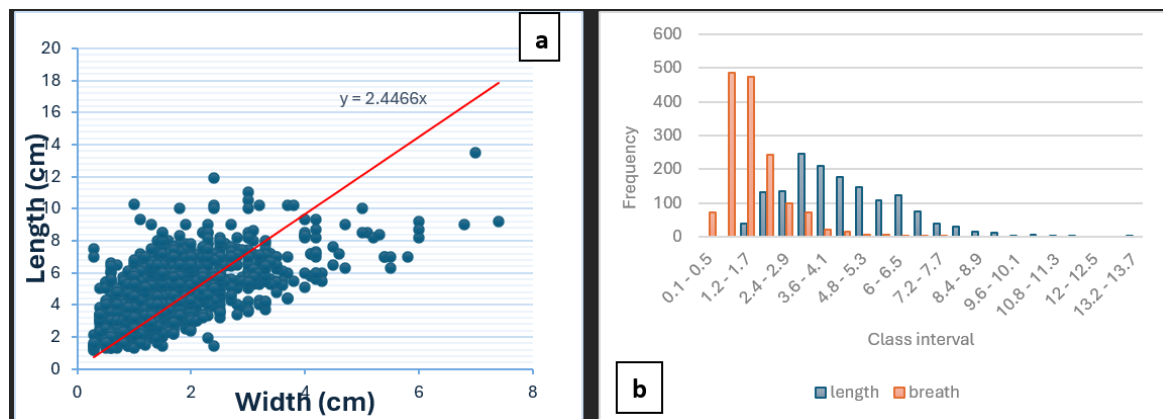


Figure 7. K-feldspar megacryst size distributions and dimensional relationships: (a) length versus width of K-feldspar megacrysts; (b) frequency distributions of megacryst length and width

The individual aspect ratios ranged from approximately 1.3 to 3.2. Therefore, the distribution was dominated by moderately elongated crystals, although a smaller proportion of crystals exhibited a greater elongation. The positive relationship between crystal length and width indicates that larger K-feldspar crystals generally have greater widths. The frequency distribution comprises multiple crystal size classes rather than a single, narrowly defined size population.

4.3 K-Feldspar Crystal Orientation

The long-axis orientations of the K-feldspar megacrysts were non-random and displayed a pronounced preferred orientation. The principal orientation population lies between approximately 040° and 060°, with a mean azimuth of approximately 054°. Therefore, the dominant orientation defines a NE–SW-trending K-feldspar fabric. A subordinate orientation population occurred between approximately 320° and 340°, corresponding to an NNW–SSE trend. The orientation distribution is shown as a rose diagram in Figure 8. The orientation of the principal strain axes defines an *X/Y* plane trending NE–SW and an elongation direction (σ_3) parallel to 054°, as shown in the figure.

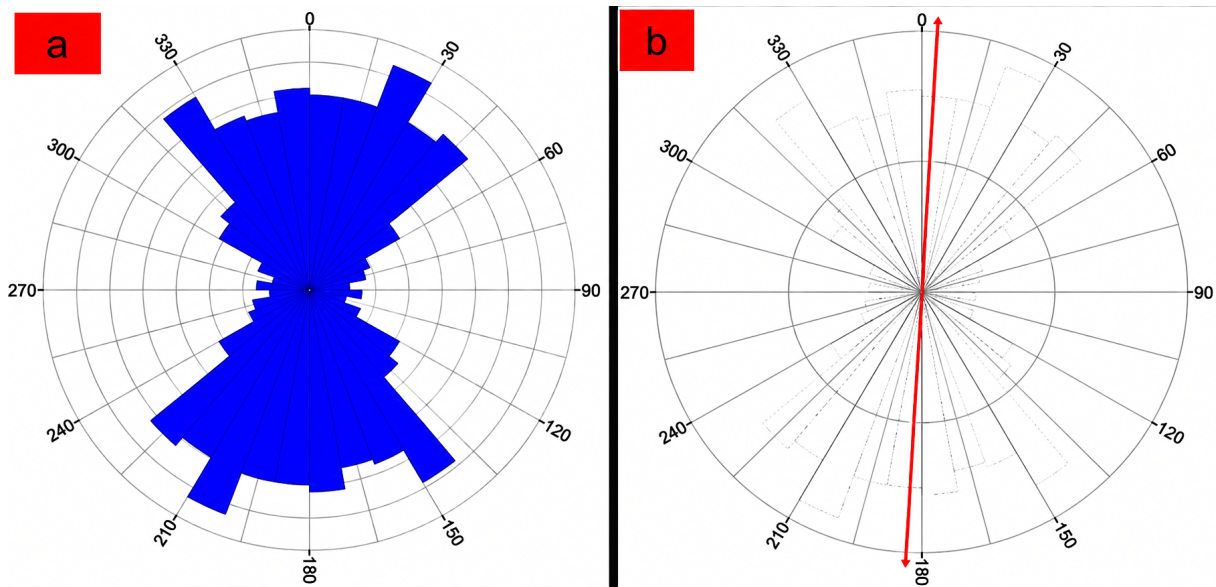


Figure 8. K-feldspar megacryst orientation and preferred fabric: (a) rose diagram of megacryst long-axis orientations showing the two principal orientation populations; (b) rose diagram showing the mean orientation, indicated by the red line, along the NNE–SSW direction

4.4 K-Feldspar Shape Anisotropy

The independently calculated shape anisotropy parameter for the K-feldspar population is $R_s = 1.575$. The mean crystal aspect ratio is $L/W = 1.95$, where R_s represents the calculated shape anisotropy parameter. Therefore, the two quantities are reported separately. The R_s value indicates measurable anisotropy in the two-dimensional shapes of the K-feldspar population. The measured crystal morphologies and orientations are shown in Figure 7 and Figure 8, respectively.

4.5 Crystal Size Distribution

The CSD analysis was performed on five representative K-feldspar populations using CSDCorrections v1.4.0. The program was used to apply stereological corrections to the two-dimensional crystal intersections and derive three-dimensional CSDs. The number of measurable crystals in each population ranged from 42 to 62 crystals per sample. Therefore, the five populations represent discrete K-feldspar crystal populations rather than a complete approximately 1,500-crystal field dataset. The CSD profiles were characterized by broadly negative relationships between the crystal population density and crystal size. Individual populations displayed differences in slope and intercept, and several profiles showed departures from simple linearity, including concave downward or hump-shaped portions of the distributions (Figure 9a and Figure 9b).

The regression parameters for the five K-feldspar populations are listed in Table 1. The regression slopes ranged from -0.414 to -0.811 mm^{-1} , whereas the intercepts ranged from -3.01 to -4.84 mm^{-4} . The combined population had a slope of -0.587 mm^{-1} and an intercept of -4.11 mm^{-4} . The variation in the regression parameters among the five populations demonstrated differences in their CSDs.

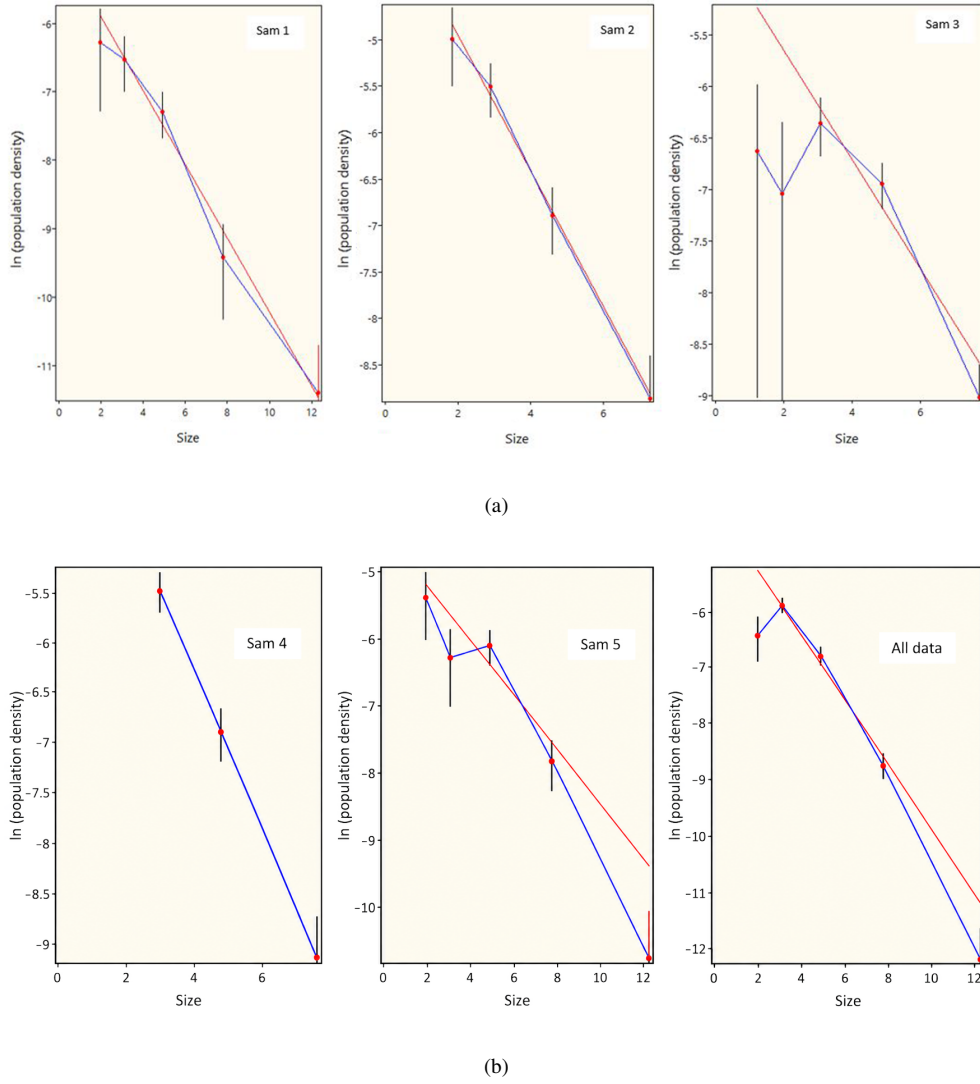


Figure 9. Crystal size distribution (CSD) diagrams of K-feldspar populations from the Abeokuta porphyritic granite: (a) samples 1–3; (b) samples 4–5 and the pooled dataset

Note: The distributions show approximately linear trends, with some profiles exhibiting concave-downward or hump-shaped patterns. Red lines indicate regression fits based on the crystal populations. Crystal size is expressed in centimeters (cm).

Table 1. Regression parameters of crystal size distributions (CSDs) for the K-feldspar populations

Population	Slope (mm^{-1})	Intercept (mm^{-4})
K-feldspar 1	−0.538	−4.84
K-feldspar 2	−0.414	−4.41
K-feldspar 3	−0.529	−4.59
K-feldspar 4	−0.811	−3.01
K-feldspar 5	−0.728	−3.51
Combined population	−0.587	−4.11

4.6 Characteristics of the Crystal Size Distribution Profiles

The five CSD profiles exhibited broadly linear negative trends over portions of their respective size ranges. However, the profiles are not identical, and several display curvature or concave downward segments. The observed curvature was particularly evident in the larger crystal-size portions of some of the distributions. The CSD distributions comprised linear and nonlinear segments, with the degree of curvature varying among the five K-feldspar populations. The CSD plots and corresponding regression relationships are shown in Figure 10.

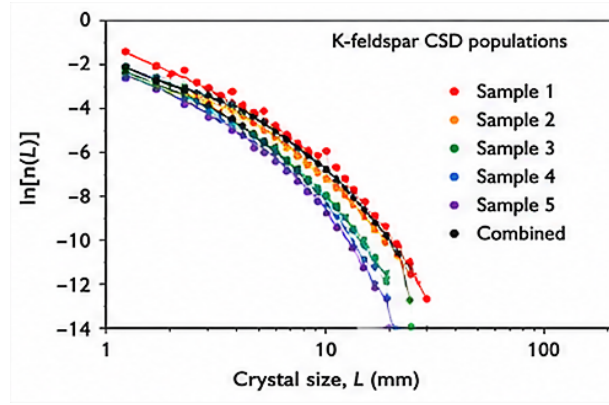


Figure 10. Stereologically corrected CSDs of the five K-feldspar populations and the combined population from the Abeokuta porphyritic granite
Note: CSD = crystal size distribution.

4.7 Regional Lineament Distribution

The remote-sensing analysis of Abeokuta SE Sheet 260 produced 1,541 lineaments, of which nine occurred within the detailed study area. The regional lineament density distribution showed greater concentrations in the northern and northeastern parts of the sheet, particularly within the basement-complex terrain. The frequency analysis identified prominent N–S and E–W lineament orientations, whereas the mean orientation of the regional lineament population trends was approximately NE–SW. Within the detailed study area, the mapped lineaments were concentrated principally around the flanks of the Ogun River.

4.8 Relationship Between K-Feldspar Fabric and Regional Lineaments

The dominant K-feldspar orientation of approximately 054° is broadly comparable to the NE–SW component of the regional lineament population. Therefore, the two datasets displayed similar directional trends. This correspondence is presented as an observed geometric relationship between the two. The significance of the relationship between magma emplacement and crystal organization is also discussed. The principal results of the quantitative investigation are listed in Table 2.

Table 2. Summary of quantitative results for the Abeokuta K-feldspar population

Parameter	Result
Number of field localities	19
Approximate K-feldspar measurements	~1,500
K-feldspar length	2.5–6.7 cm
K-feldspar width	1.2–3.5 cm
Mean length	4.56 cm
Mean width	2.34 cm
Aspect-ratio range	1.3–3.2
Mean (L/W)	1.95
Shape anisotropy (R_s)	1.575
Dominant orientation	054°
Dominant fabric	NE–SW
Secondary orientation	320°–340°
CSD populations	5
CSD population size	42–62 crystals/sample
CSD slope	–0.414 to –0.811 mm ^{–1}
CSD intercept	–3.01 to –4.84 mm ^{–4}
Combined CSD slope	–0.587 mm ^{–1}
Combined CSD intercept	–4.11 mm ^{–4}
Regional lineaments	1,541
Lineaments within the detailed study area	9

Note: CSD = crystal size distribution; L = long-axis length; W = short-axis width.

Overall, the results demonstrate that the Abeokuta porphyritic granite contains a measurable and directionally

organized K-feldspar megacryst population with a preferred orientation. The crystals exhibited moderate morphological anisotropy, a dominant NE–SW long-axis orientation, and variable CSDs. The regional lineament population also contained a significant NE–SW component.

5 Discussion

5.1 K-Feldspar Megacryst Morphology and Crystallization

The Abeokuta porphyritic granite contains abundant euhedral to subhedral K-feldspar megacrysts in a medium-to-coarse-grained quartz–feldspar–biotite matrix. The megacrysts ranged from approximately 2.5 to 6.7 cm in length and 1.2 to 3.5 cm in width, with mean dimensions of approximately 4.56 and 2.34 cm, respectively. The mean length-to-width ratio of 1.95 indicates moderately elongated crystals. Perthitic intergrowths and localized compositional zoning indicate that K-feldspar crystallization occurred during progressive evolution of the granitic melt [3, 4]. Large K-feldspar crystals in granitic systems can result from prolonged growth in mobile magma, late-stage textural coarsening, or changes in physicochemical conditions governing nucleation and growth. Thus, K-feldspar megacrysts may record composite crystallization rather than a single stage of magma evolution. In the Abeokuta granite, variations in crystal size and morphology are consistent with sustained growth under changing magmatic conditions [3, 22].

The long axes of the K-feldspar megacrysts showed a pronounced preferred orientation mainly between 040° and 060°, with a mean azimuth of approximately 054°. This defines a dominant NE–SW-trending crystal fabric, with a weaker NNW–SSE orientation, indicating that the crystals were not randomly distributed. Such alignment can develop through rotation and mechanical organization within flowing magma while the melt remains sufficiently mobile. Crystal orientation can therefore preserve information about magma flow and emplacement without necessarily recording post-solidification tectonic deformation [2, 5]. The observed K-feldspar fabric is primarily interpreted as magmatic or synmagmatic, consistent with conspicuous megacrysts in a granular igneous matrix and limited petrographic evidence of solid-state deformation. However, preferred fabric does not preclude tectonic influence; regional deformation may have influenced magma ascent, emplacement, and flow while crystal alignment developed during the magmatic stage [23]. Accordingly, the 054° orientation is treated as the measured crystallographic or long-axis fabric direction rather than a direct measurement of a principal stress axis.

5.2 Shape Anisotropy and Deformation

The corrected shape anisotropy parameter of the K-feldspar population was $R_s = 1.575$. This value indicates measurable anisotropy of observed crystal shapes. The anisotropy may reflect primary crystal habit, growth kinetics, magmatic flow, and limited modification during emplacement or cooling. Moderate crystal elongation, coherent preferred orientation, and limited petrographic evidence for solid-state deformation are more consistent with a predominantly magmatic origin for the fabric than with intense post-solidification deformation. The K-feldspar population may have acquired its shape and orientation while magma was mobile enough for crystal rotation and organization, followed by crystallization and cooling that preserved the fabric [2, 24]. This interpretation reconciles moderate shape anisotropy with weak solid-state deformation in the granite, as significant crystal organization does not require substantial deformation after crystallization. Regional lineament analysis identified a heterogeneous basement terrain with prominent N–S, E–W, and NE–SW trends. The structural framework included a significant NE–SW component broadly comparable to the 054° orientation of the K-feldspar megacrysts [8, 25]. Correspondence between regional trends and the K-feldspar fabric suggests that structural anisotropy may have influenced emplacement and flow of the Abeokuta magma [26]. Pre-existing weaknesses could have facilitated magma ascent and emplacement.

5.3 Crystal Size Distribution and Crystallization Dynamics

The K-feldspar CSD analysis provides an independent constraint on crystal population evolution. The five analyzed populations contained 42–62 measurable crystals per sample and yielded regression slopes from -0.414 to -0.811 mm^{-1} , with a combined slope of -0.587 mm^{-1} . Corresponding intercepts ranged from -3.01 to -4.84 mm^{-4} , with a combined intercept of -4.11 mm^{-4} . Variation in slope and intercept among the five populations indicates that K-feldspar crystals did not develop under a single, invariant crystallization regime [27]. Differences in CSD parameters can reflect changes in nucleation and growth rates, crystal accumulation or removal, melt conditions, residence time, or textural coarsening. The variation suggests that Abeokuta granite crystallization was dynamic, not a single steady-state process. Negative CSD relationships indicate more small than large crystals within analyzed populations, as expected for magmatic systems. Departures from linearity and differences among populations indicate progressive, non-uniform crystal population evolution and prolonged crystal growth during magmatic history [28].

5.4 Evidence for Textural Coarsening

Textural coarsening explains the development and modification of the K-feldspar population. During crystallization, larger crystals may grow at the expense of smaller ones through changes in interfacial energy,

diffusion, and melt availability. These processes can modify established crystal populations and produce large crystals without requiring megacrysts to form during the crystallization stages [3, 4]. The CSD characteristics of the Abeokuta K-feldspar population are compatible with this modification. However, the CSD data do not show that coarsening alone formed the megacrysts. Nucleation, growth, accumulation, resorption, and melt conditions may also shape the CSD. Thus, the megacrysts reflect crystal growth and textural modification during magma evolution, with coarsening as a plausible component, consistent with established CSD studies and extending the approach to the Abeokuta porphyritic granite.

5.5 Crystal Size Distribution Variation and Implications for Magma Evolution

The differences among the five CSD populations suggest that the crystallizing system experienced variable conditions. This variation could reflect differences in cooling rate, melt fraction, crystal concentration, nucleation density, or residence time within pluton [28, 29]. Curved CSDs indicate departure from an exponential crystal-size relationship, but this curvature is not diagnostic of a single process. Although magma or mixing and recharge can produce complex CSD patterns [30], similar features can result from accumulation, removal, coarsening, or changing nucleation and growth conditions. Thus, the CSD data indicate changing crystal population dynamics, not evidence of mixing or recharge.

5.6 Regional Petrogenetic Context

Published geochemical studies of Pan-African granitoids in southwestern Nigeria provide a broader context for understanding the possible evolution of the Abeokuta granite. Regional granitoids have been associated with processes such as fractional crystallization, crustal assimilation, and, in some bodies, interactions between crustal and mantle-derived magmas [17, 31, 32]. These regional observations are relevant because they demonstrate the importance of open-system magmatic processes during the development of Pan-African granitoids in southwestern Nigeria. They also provide plausible mechanisms for the variations in crystal populations and textural characteristics. These characteristics from the regional context could be extended to imply that the magma responsible for the Abeokuta granite likely was derived from a hybrid source involving the partial melting of crustal materials with limited mantle input. Such hybridization processes [33] are known to promote textural coarsening and crystal size variability, consistent with the concave-downward trends observed in the CSD plots (Figures 9a and Figure 9b).

The petrographic characters displayed by the porphyritic granite are consistent with those of granitoids of a magmatic origin that crystallized under fluctuating physicochemical conditions [34]. The alignment of these megacrysts and the preferred NE–SW orientation of strain ellipsoids (Figure 8) suggest that crystallization occurred contemporaneously with deformation. These features are comparable to those documented in other Pan-African granitoids across southwestern Nigeria [7], where magmatic fabrics developed during emplacement within a transpressional regime.

5.7 Implications for Emplacement During the Pan-African Tectonomagmatic Event

The combined observations of structural features, crystal morphometry, and CSDs provide evidence that granite crystallized and developed its internal fabric during a tectonically active period of the Pan-African Orogeny. The NE–SW K-feldspar fabric, its correspondence with regional structural trends, and the limited evidence for subsequent solid-state deformation suggest that a substantial component of the observed fabric developed while the magma was still mobile. Therefore, granite records a close relationship between magma emplacement, crystal organization, and progressive crystallization.

A plausible emplacement sequence involves magma ascent through structurally favorable zones, development of flow within the emplaced magma, preferential organization of K-feldspar crystals, continued crystal growth and textural adjustment, followed by progressive cooling and solidification. This model does not require the crystal fabric to represent a direct tectonic strain ellipsoid. Instead, tectonic structures may have controlled the emplacement environment, while magmatic flow generated the measurable K-feldspar orientation. Therefore, the Abeokuta granite can be regarded as a record of synmagmatic interaction between regional structural control and magmatic crystallization, while the precise kinematic relationship between emplacement and regional deformation remains unresolved. This study hence proposes a multistage model for the evolution of the K-feldspar population. During the early stages of crystallization, K-feldspar nucleated and grew within a progressively crystallizing granitic melt. Continued growth under changing physicochemical conditions produces the large euhedral to subhedral megacrysts observed today. As crystallization progressed, the magma retained sufficient mobility for the growing crystals to become preferentially organized. Crystal rotation and mechanical interactions within the flowing melt produced the dominant NE–SW orientation centered at approximately 054°. Subsequent crystallization reduced magma mobility and increasingly preserved the established crystal fabric. Simultaneously, changes in nucleation, growth, and crystal-population dynamics produced the observed variability in CSD characteristics. Textural coarsening may have contributed to the development of the largest K-feldspar crystals during the later stages of crystallization. The

resulting granite therefore preserves two complementary aspects of its magmatic history: the preferred orientation of K-feldspar records crystal organization during emplacement, whereas the CSD records changes in the crystal population during progressive crystallization and cooling of the granite.

The proposed evolutionary sequence of the Abeokuta porphyritic granite, integrating regional structural control, magma emplacement, K-feldspar crystallization and organization, and subsequent crystal-population evolution, is summarized in Figure 11. The conceptual model emphasizes the interaction between regional structural control, magmatic flow, K-feldspar crystallization, and crystal population evolution.

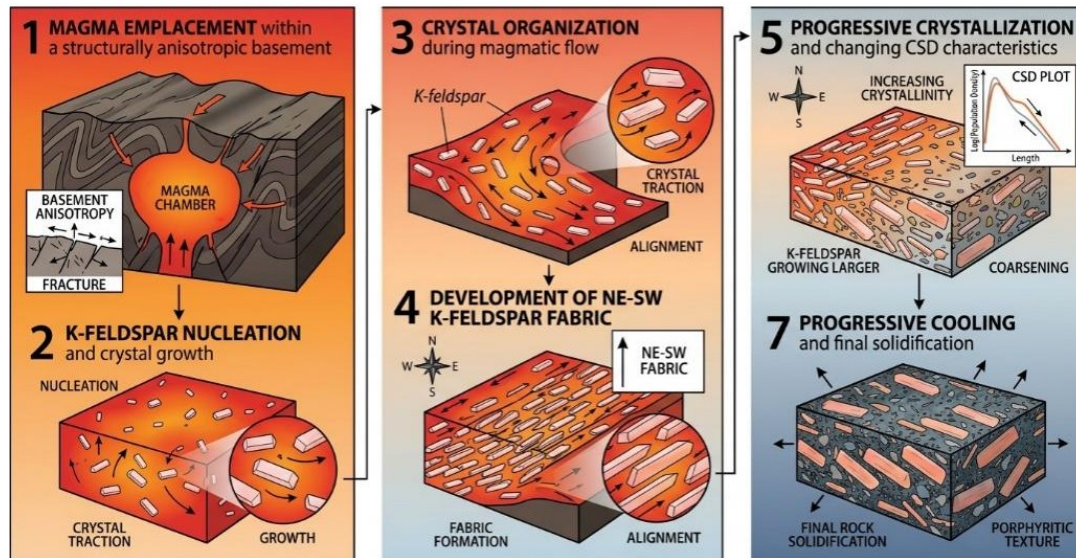


Figure 11. Conceptual model depicting the evolutionary sequence for the Abeokuta porphyritic granite

Note: CSD = crystal size distribution.

The results demonstrate the value of combining crystal morphometry, preferred orientation analysis, CSD, and regional structural mapping in the investigation of porphyritic granitoids. K-feldspar megacrysts are particularly useful because their size, morphology, and orientation preserve information about the physical conditions prevailing during the magma emplacement and crystallization. The Abeokuta granite illustrates that a strong magmatic fabric does not necessarily imply an intense post-crystallization deformation. Similarly, complex CSD patterns do not uniquely identify magma mixing and recharge. Instead, these datasets are most informative when interpreted together with petrographic observations and regional-structural information.

5.8 Limitations and Implications for Future Work

The principal limitation of the present interpretation is that field morphometric measurements provide two-dimensional observations of three-dimensional K-feldspar crystals. Although stereological corrections were applied during the CSD analysis, the relatively small CSD populations, comprising 42–62 crystals per sample, further indicate that the CSD results should be regarded as representative of the analyzed populations rather than exhaustive of the entire pluton. Finally, the lack of new whole-rock geochemical and isotopic data limits the interpretation of the magma source and open-system processes. Future investigations combining whole-rock geochemistry, mineral chemistry, zircon uranium–lead (U–Pb) geochronology, zircon lutetium–hafnium (Lu–Hf) isotopes, crystallographic preferred-orientation analysis, and three-dimensional imaging would provide stronger constraints on the relationship between magmatic crystallization, structural emplacement, and subsequent deformation.

6 Conclusions and Implications

6.1 Conclusions

This study investigated the morphology, preferred orientation, and CSD of K-feldspar megacrysts in the Abeokuta porphyritic granite in southwestern Nigeria to constrain the crystallization and emplacement history of the pluton within the Pan-African tectonomagmatic framework. The K-feldspar megacrysts were predominantly euhedral to subhedral and ranged from approximately 2.5 to 6.7 cm in length and 1.2 to 3.5 cm in width, with mean dimensions of 4.56 cm and 2.34 cm, respectively. The mean length-to-width ratio of 1.95 indicates a moderately elongated crystal population. The corrected shape anisotropy parameter ($R_s = 1.575$) further demonstrates measurable two-dimensional anisotropy within the K-feldspar population. The long axes of the K-feldspar megacrysts exhibited a pronounced preferred orientation centered at approximately 054° , defining a dominant NE–SW-trending magmatic

fabric. The coherent orientation of the megacrysts indicates that the crystals were preferentially organized rather than randomly distributed. In conjunction with the petrographic characteristics and limited evidence for solid-state deformation, the fabric is interpreted primarily as a synmagmatic feature developed while the crystallizing magma retained sufficient mobility for crystal organization.

The CSD analysis provides additional constraints on the evolution of the K-feldspar population. Across the five analyzed populations, each sample contained 42–62 measurable crystals. Regression analysis produced slopes ranging from -0.414 to -0.811 mm^{-1} , while the overall slope was -0.587 mm^{-1} . The intercept values varied from -3.01 to -4.84 mm^{-4} , with an overall intercept of -4.11 mm^{-4} . The variation in CSD parameters and departures from simple linearity indicate that the K-feldspar population developed under changing crystallization conditions rather than through a single, invariant nucleation and growth regime. The combined morphometric and CSD evidence is consistent with progressive K-feldspar growth, accompanied by changing crystal-population dynamics and possible textural coarsening during granite evolution. However, CSD characteristics do not uniquely identify a single process, such as magma mixing, recharge, or coarsening. These mechanisms remain possible explanations rather than proven processes. Regional lineament analysis identified a significant NE–SW component within the structural framework of the Abeokuta area. The broad correspondence between this regional trend and the 054° K-feldspar fabric suggests that structural anisotropy may have influenced the magma emplacement and flow. However, the available data do not independently establish the orientations of the principal stress axes or a specific regional kinematic regime.

Overall, the Abeokuta porphyritic granite is interpreted as a Pan-African granitoid, whose K-feldspar population preserves complementary records of crystallization, magmatic crystal organization, and progressive textural evolution. The preferred K-feldspar fabric primarily records synmagmatic organization during emplacement, whereas the CSD characteristics record changes in crystal-population evolution during progressive crystallization and cooling.

6.2 Implications

The results demonstrate that K-feldspar megacrysts constrain the internal dynamics of porphyritic granitoids. Their preferred orientation indicates crystal organization within magma, while morphology and CSD provide complementary information on crystal growth and population evolution. The NE–SW orientation of the K-feldspar population, together with the regional structural framework, suggests that emplacement was influenced by the host basement rock architecture. This supports a model in which pre-existing or contemporaneous crustal structures provide pathways or anisotropies for magma ascent and emplacement. This study also demonstrates that a well-developed K-feldspar fabric does not necessarily require intense post-solidification deformation. Crystal alignment may develop while magma remains mobile enough for rotation, interaction, and preferential organization of crystals. Therefore, a preferred K-feldspar orientation in porphyritic granite should be evaluated with petrographic evidence of deformation before interpretation as a tectonic strain fabric. CSD results show the value of quantitative crystal population analysis in K-feldspar megacryst-bearing granitoids. Differences in slope, intercept, and profile geometry reveal crystal population behavior not apparent from conventional petrography alone when integrated with morphology and orientation. The Abeokuta findings constrain growth and emplacement within the Pan-African granitoid assemblage of southwestern Nigeria, but resolving the magma source requires geochemical and isotopic evidence from the granite itself.

6.3 Overall Conclusion

This study quantitatively analyzed approximately 1,500 K-feldspar megacrysts from 19 localities to characterize their morphology, preferred orientation, and CSD. The megacrysts exhibit a pronounced 054° NE–SW preferred orientation, defining a coherent K-feldspar fabric that records systematic crystal organization during magma emplacement. CSD patterns reveal non-uniform crystallization and variable crystal-growth dynamics, while the correspondence between the megacryst fabric and regional structural anisotropy suggests that pre-existing structures likely influenced magma ascent and emplacement. Collectively, the integrated CSD, morphometric, and fabric analyses provide quantitative constraints on the synmagmatic emplacement and evolution of the Abeokuta porphyritic granite.

The Abeokuta porphyritic granite preserves a quantitatively measurable K-feldspar fabric and a variable CSD that together provide evidence for a dynamically evolving magmatic system. The moderate crystal anisotropy, coherent 054° NE–SW orientation, and variable CSD characteristics indicate that K-feldspar crystallization and organization occurred during progressive magma evolution and emplacement. The results are most consistent with a predominantly synmagmatic origin for the observed crystal fabric, potentially influenced by the regional structural architecture of the southwestern Nigerian Basement Complex. This study consequently demonstrates that the integration of K-feldspar morphometry, fabric analysis, and CSD can provide valuable constraints on granitoid emplacement and crystallization history, while direct geochemical and isotopic investigation remains necessary to resolve the source and open-system evolution of the Abeokuta magma.

Author Contributions

Conceptualization, E.E.I.; methodology, T.I.S.; software, T.I.S.; validation, E.E.I. and T.I.S.; formal analysis, T.I.S.; investigation, E.E.I. and T.I.S.; resources, E.E.I. and T.I.S.; data curation, T.I.S.; writing—original draft preparation, T.I.S.; writing—review and editing, E.E.I.; visualization, E.E.I. and T.I.S.; supervision, E.E.I.; project administration, E.E.I. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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